AMIT VIKRAM REX

Curriculum Vitae

Mechanics of Material Lab
Department of Mechanical Engineering
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Experience

Oct 2025–Present **Post-Doctoral Associate, Mechanical Engineering**, *Indian Institute of Technology Guwahati*, Assam.
Hydrogen Assisted Cracking
Advisor: Dr. Nipal Deka

Feb 2025–Oct 2025 **Project Fellow, Mechanical Engineering**, *Indian Institute of Technology Patna*, Bihar.
Hygrothermal Fracture Analysis of Carbon Fiber Reinforced Polymer
Advisor: Dr. Sunil Kumar Singh

Education

2019–2025 **PhD, Mechanical Engineering**, *Indian Institute of Technology Patna*, Bihar.
Finite Element Analysis, Notch fatigue simulation, Fatigue and fracture mechanics (Low-cycle fatigue, High-cycle fatigue, Fatigue crack propagation), Cyclic plastic deformation modeling

2016–2018 **Master of Technology, Applied Mechanics**, *Motilal Nehru National Institute of Technology Allahabad*, Prayagraj, Uttar Pradesh.

2012–2016 **Bachelor of Technology, Mechanical Engineering**, *Dehradun Institute of Technology, Uttarakhand Technical University*, Dehradun, Uttarakhand.

2009–2011 **Intermediate Examination**, *K.R.K Maudal School*, Lakhisarai, Bihar.

2008–2009 **Secondary School Examination**, *Imperial Public School*, Gopalganj, Bihar.

Research Interest

Physics-informed modeling, Machine/deep learning approaches, Path-dependent material behavior, Fatigue and fracture mechanics, Hydrogen-assisted cracking.

Publications

Journal Articles

2025 **Amit Vikram Rex, Surajit Kumar Paul, Akhilendra Singh**, Effects of equi-biaxial and uni-axial tensile pre-straining paths on fatigue crack propagation behaviour of 2024-T4 aluminium alloy, *Materials Today Communications, 49, 114062.*

2024 **Amit Vikram Rex, Surajit Kumar Paul, Akhilendra Singh**, Assessing notch fatigue performance of aluminium alloy AA2024-T4 after uni-axial and equi-biaxial tensile pre-straining: Experimental observations and predictive models, *Theoretical and Applied Fracture Mechanics, 131, 104402.*

2023 **Amit Vikram Rex, Surajit Kumar Paul, Akhilendra Singh**, Influence of uniaxial and equi-biaxial tensile pre-straining on the high cycle and notch fatigue behaviour of AA2024-T4 aluminium alloy, *Theoretical and Applied Fracture Mechanics, 128, 104176.*

2023 **Amit Vikram Rex, Surajit Kumar Paul, Akhilendra Singh**, The influence of equi-biaxial and uniaxial tensile pre-strain on the low cycle fatigue performance of the AA2024-T4 aluminium alloy, *International Journal of Fatigue, 173, 107699.*

2023 PP Dubey, **Amit Vikram Rex,** Abhishek Raj, Surajit Kumar Paul, Influence of pre-strain on tensile response of extra deep drawing (EDD) steel under varying strain rates and crash performance, *Journal of Alloys and Metallurgical Systems, 4, 100036.*

Communicated Journal Article

2026 Pranaw Parihar, **Amit Vikram Rex,** Sunil Kumar Singh, Crack Propagation Analysis of CFRP Composites using DIC and Strain Energy Density based XIGA, *Theoretical and Applied Fracture Mechanics* (Under Review).

Conference Proceedings

2018 **Amit Vikram Rex,** Ashutosh Kumar Upadhyay, Mechanical Behavior of Polyurethane Foam under Quasi-Static and Repeated Impact Loading, *First Symposium and Workshop for Analytical Youth on Applied Mechanics (SWAYAM 2018), BITS Pilani, KK Birla Goa Capmus,pp.11-12,07/2018,Published By. Ane Books Pvt. Ltd.*

Research Experience

Indian Institute of Technology, Patna

Assessing the effects of equi-biaxial and uni-axial tensile pre-straining paths on fatigue crack propagation behaviour of AA2024-T4 aluminium alloy.
Investigated the effect of tensile pre-straining on tensile properties, fracture resistance, and fatigue crack propagation using experiments and XFEM simulations. Observed minimal change in FCP rate, slight improvement in fatigue life, and a 3% increase in conditional fracture toughness (KQ).

Assessing notch fatigue performance of AA2024-T4 aluminium alloy after uni-axial and equi-biaxial tensile pre-straining.
Studied the influence of pre-straining paths on notch fatigue behaviour using a physics-based approach. Combined high-cycle fatigue experiments with the Theory of Critical Distance (TCD) and cyclic plastic zone (CPZ) concept to accurately predict fatigue life.

Analyzing the effect of tensile pre-straining on fatigue behaviour of AA2024-T4 aluminium alloy.
Evaluated the impact of uni-axial and equi-biaxial pre-straining on tensile and fatigue behaviour. EBSD analysis showed increased lattice distortion, while pre-straining enhanced tensile strength, yield stress, and fatigue strength. TCD predicted fatigue life within a ±2 scatter band.

Assessing the effect of tensile pre-strain on the low-cycle fatigue performance of AA2024-T4 aluminium alloy.
Examined low-cycle fatigue behaviour under different pre-straining conditions by comparing fatigue life and cyclic stress–strain response. Uniaxial pre-straining improved fatigue life, while equi-biaxial pre-straining showed comparable performance to unstrained material.

Assessing the influence of pre-strain on tensile response and crash performance of extra deep drawing steel at different strain rates.
Analyzed the effect of pre-strain on tensile behaviour of EDD steel across strain rates. Observed increased strength and energy absorption with reduced elongation, and used a validated model to predict crash performance.

Motilal Nehru National Institute of Technology Allahabad, Prayagraj

Experimental study on the dynamic response of corrugated core sheet subjected to low-velocity impact.
Investigated low-velocity impact behaviour of corrugated sheets through dynamic and quasi-static experiments. Evaluated responses of filled and unfilled structures, including mechanical properties of polyurethane foam core.

Fellowships & Awards

2019 –2024 ***Institute Fellowship*** as a PhD research scholar at the Indian Institute of Technology Patna.

2016 –2018 ***GATE Scholarship*** of Ministry of Electronics and Information Technology (MeitY), Government of India, as an M.Tech student at the Motilal Nehru National Institute of Technology Allahabad.

Computer skills

FEA Tools: Abaqus and ANSYS APDL

CAD Design Software: Solidworks

Scientific Computing: Python, PyTorch

Position of Responsibility

2017 **Volunteer, INCAM 2017**, MNNIT Allahabad.

Teaching Assistantship

Fall 2019 **ME110: Workshop**, IIT Patna.

Spring 2020, Spring 2021 **ME292: Measurement Laboratory**, IIT Patna.

Fall 2020 **ME231: Engineering Materials Lab**, IIT Patna.

Fall 2022 **ME313: Design of Machine Elements Lab**, IIT Patna.

Fall 2022, Fall, 2023 **ME533: Finite Element Analysis Lab**, IIT Patna.

Fall 2018 **AM1153: Engineering Mechanics Lab**, MNNIT Allahabad.

Referees

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